

Machine Learning

Trainer: Prashant N.

Supervised Learning } applicable for the dataset that has both features and label

Regression

Features: any type of data
Label: Continuous ND

Classification

Features: any type of data
Label: Categorical / Discrete ND.

Inferential Stat modelling process

Assuming the given dataset is COMPATIBLE for ML algo.

- ① Data must be COMPLETE
- ② Data must be STRICTLY NUMERIC

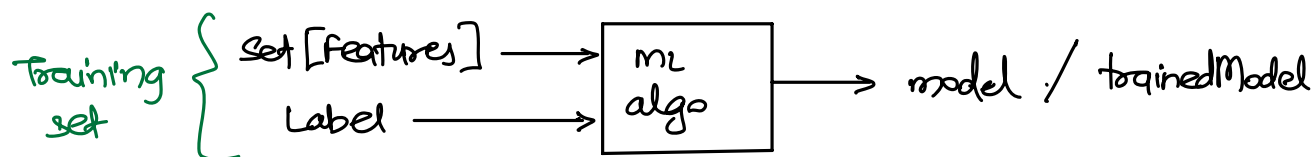
① Separate data as features and label (represent the data in the form of numpy array)

② Split the data into Training set and Testing set.

Training set → used to train the model (Data Convergence)

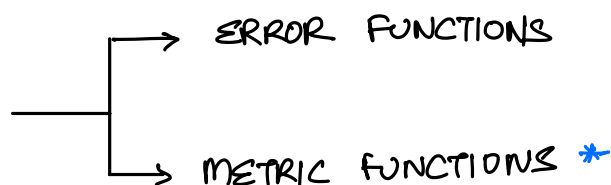
Testing set → used to check the quality of the model.

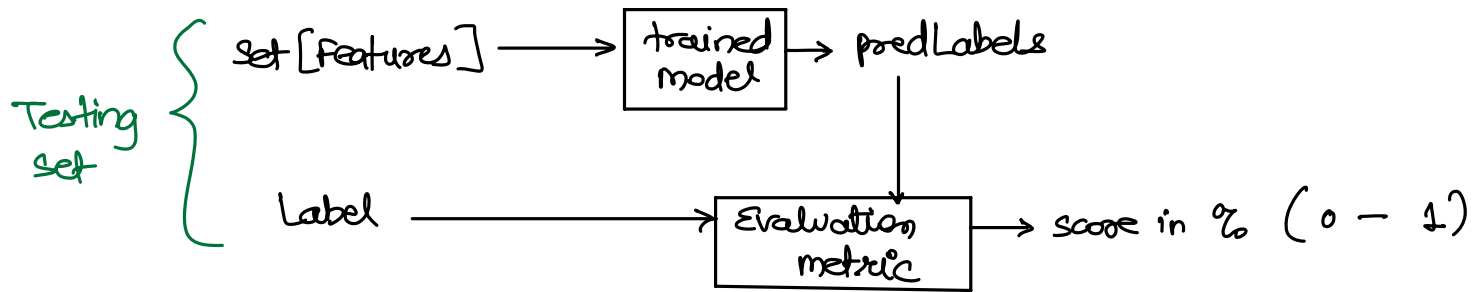
③ Select the relevant ML algorithms and start the training process.



④ Check the quality of the model / model's prediction.

Goal: check the quality of the model w.r.t prediction





⑤ Decide whether to Approve model or Improve model.

if eval passed:

Approve the model and deploy the same

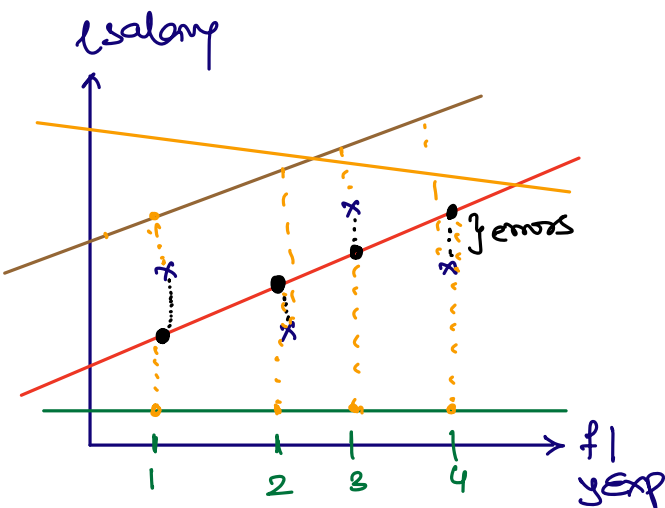
else:

Improve model and recheck the quality

Linear Regression

This algorithm is inspired from SLOPE INTERCEPT FORMULA.

$$y = mx + c.$$



line of best fit

Goal: Create a model that can predict the salary of the employee based on employee's years of experience.